

Classification Models and Adult Census Income Dataset

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Introduction

This study explores predictive features for classifying adult income using Sci-Kit Learn classification models, to identify demographic and socioeconomic variables associated with earning an income higher than or equal to 50K or less than 50K. The findings have implications for understanding which features are most influential in predicting income level.

The “Adult Census Income” dataset from the UC Irvine Machine Learning Repository was used for this analysis. This dataset is widely applied in classification tasks (Becker & Kohavi, 1996), making it a standard benchmark for evaluating classification models. The target variable is binary, indicating whether an individual’s income exceeds 50K or is at or below 50K annually.

Methodology

The original dataset consists of separate training and test files containing 14 predictor variables (see Table 1). For this analysis, both datasets were combined into a single dataset of 48,842 rows to allow for a controlled train-test split during preprocessing. Before modeling, two variables were removed: `fnlwgt`, since it is a sampling weight rather than a meaningful feature, and `education_num`, as it is redundant with the categorical education variable. The categorical version was retained for improved interpretability.

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
age	Feature	Integer	Age	N/A		no
workclass	Feature	Categorical	Income	Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.		yes
fnlwgt	Feature	Integer				no
education	Feature	Categorical	Education Level	Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.		no
education-num	Feature	Integer	Education Level			no
marital-status	Feature	Categorical	Other	Married-civ-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.		no
occupation	Feature	Categorical	Other	Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.		yes

relationship	Feature	Categorical	Other	Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.		no
race	Feature	Categorical	Race	White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.		no
sex	Feature	Binary	Sex	Female, Male.		no
capital-gain	Feature	Integer				no
capital-loss	Feature	Integer				no
hours-per-week	Feature	Integer				no
native-country	Feature	Categorical	Other	United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinidad&Tobago, Peru, Hong, Holand-Netherlands.		yes
income	Target	Binary	Income	>50K, <=50K.		no

Table 1: Variables Table from UCI Adult Census Income

The dataset exhibits class imbalance, with an approximate 3:1 ratio between the two classes [Figure 1]. Numeric features showed no strong multicollinearity and, therefore, dimensionality reduction techniques were not applied [Figure 2]. Categorical variables were transformed using one-hot encoding.

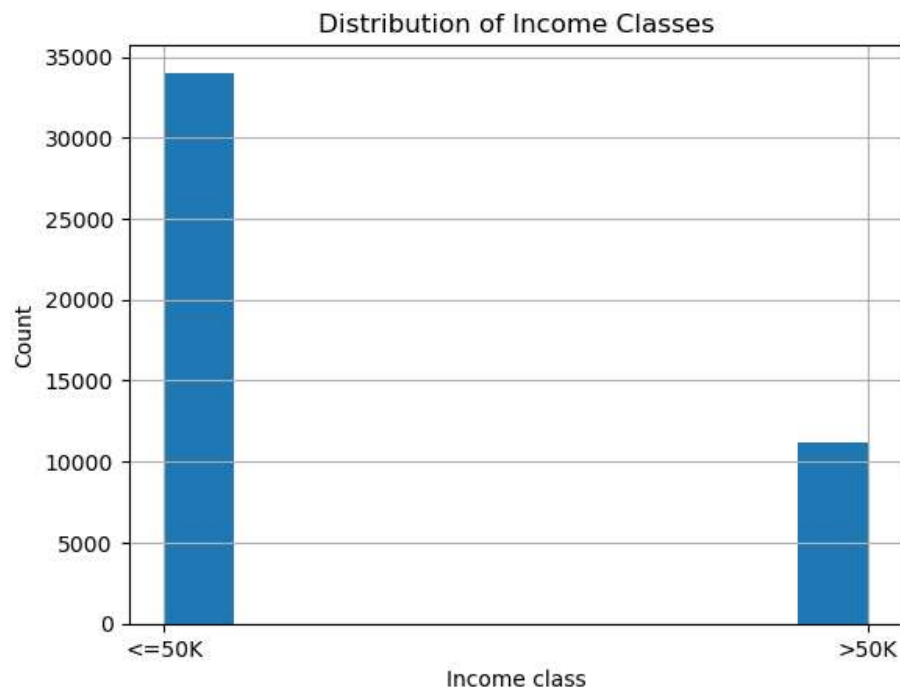


Figure 1: Distribution of Income Class

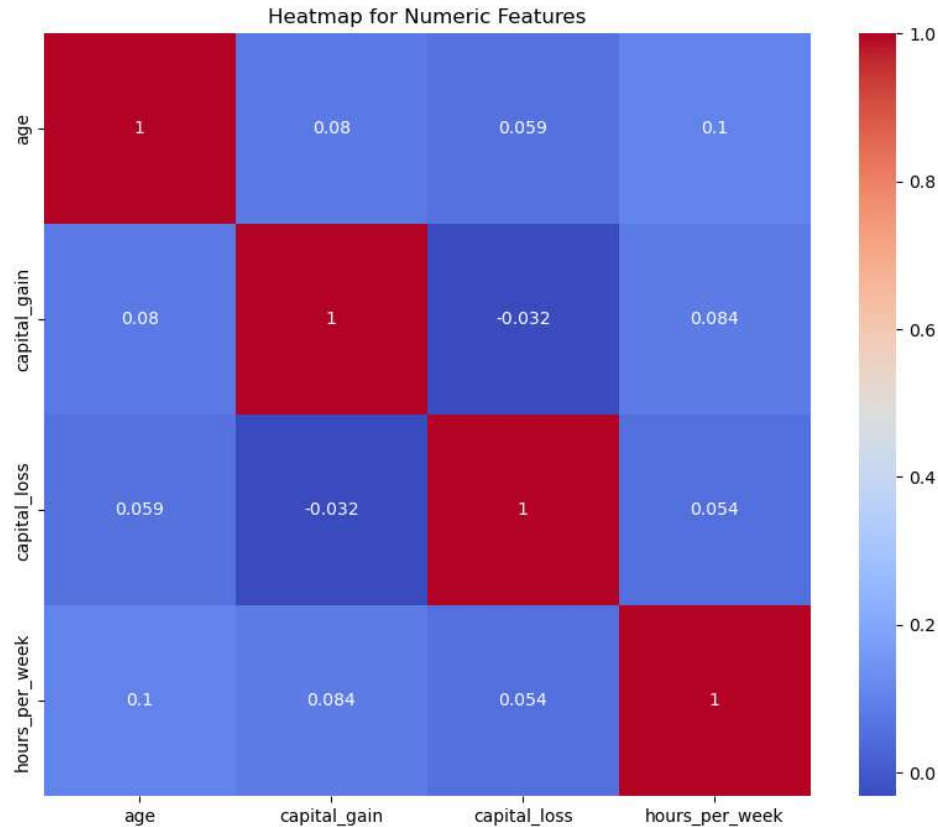


Figure 2: Heatmap (Correlation) of Numeric Features

Data cleaning included handling missing or invalid entries, where “?” values were replaced with null values and removed. Additionally, formatting inconsistencies in the target variable were corrected. The final cleaned dataset contained 45,222 instances.

The income label was also one-hot encoded, with income higher than or equal to 50K defined as the positive class. Model performance was evaluated using confusion matrices, accuracy, precision, recall, F1-score, and ROC-AUC, with particular emphasis on ROC-AUC due to the class imbalance. ROC-AUC was selected as a primary evaluation metric because it is more robust in imbalanced classification settings, where accuracy alone may be misleading.

The classification models employed in this study include `DecisionTreeClassifier`, `LogisticRegression`, and `K-Nearest Neighbors`, implemented using the Sci-Kit Learn library. In addition, a `RandomForestClassifier` was used as an ensemble method to provide a performance

comparison against the individual models and to evaluate whether ensemble learning improves predictive performance on the imbalanced dataset.

To optimize model performance, GridSearchCV was applied to each classifier to identify the best hyperparameters. For tree-based models, parameters such as maximum depth were tuned; for K-Nearest Neighbors, the number of neighbors and distance metrics were optimized; and for Logistic Regression, regularization strength and solver-related parameters were adjusted. The optimal hyperparameter configurations selected included a maximum depth of 10 for the DecisionTreeClassifier. For Logistic Regression, a regularization strength of 0.1 with an L2 penalty was chosen. For K-Nearest Neighbors, the optimal hyperparameters used 7 nearest neighbors with Manhattan distance as the distance metric [Table 2].

	Model	Parameters	Training Time (sec)	Testing Time (sec)	Accuracy	Precision	Recall	F1_score	ROC-AUC
0	DecisionTreeClassifier	{'ccp_alpha': 0.0, 'class_weight': None, 'criterion': 'gini', 'max_depth': 10, 'max_features': None, 'max_leaf_nodes': None, 'min_impurity_decrease': 0.0, 'min_samples_leaf': 1, 'min_samples_split': 2, 'min_weight_fraction_leaf': 0.0, 'monotonic_cst': None, 'random_state': 42, 'splitter': 'best'}	0.2021	0.0044	0.85	0.85	0.85	0.85	0.89
1	LogisticRegression	{'C': 0.1, 'class_weight': None, 'dual': False, 'fit_intercept': True, 'intercept_scaling': 1, 'l1_ratio': None, 'max_iter': 200, 'multi_class': 'deprecated', 'n_jobs': None, 'penalty': 'l2', 'random_state': 42, 'solver': 'lbfgs', 'tol': 0.0001, 'verbose': 0, 'warm_start': False}	0.2561	0.002	0.84	0.84	0.84	0.84	0.9
2	KNeighborsClassifier	{'algorithm': 'auto', 'leaf_size': 30, 'metric': 'manhattan', 'metric_params': None, 'n_jobs': None, 'n_neighbors': 7, 'p': 2, 'weights': 'uniform'}	0.0198	8.1469	0.83	0.82	0.83	0.82	0.87

Table 2: Classification Results and Optimal Hyperparameters

Results and Observations

The classification models performed similarly in predicting the income label based on their ROC-AUC scores. ROC-AUC was used as the primary evaluation metric due to its suitability for imbalanced classification problems, as it measures the true positive and false positive rates across thresholds.

However, the models differed in the features they identified as most important. Decision Tree and Logistic Regression, for example, highlighted different predictors of income, as shown in Figure 3 and 4. The Decision Tree model identified being married as an important feature associated with earning an income greater than or equal to 50K.

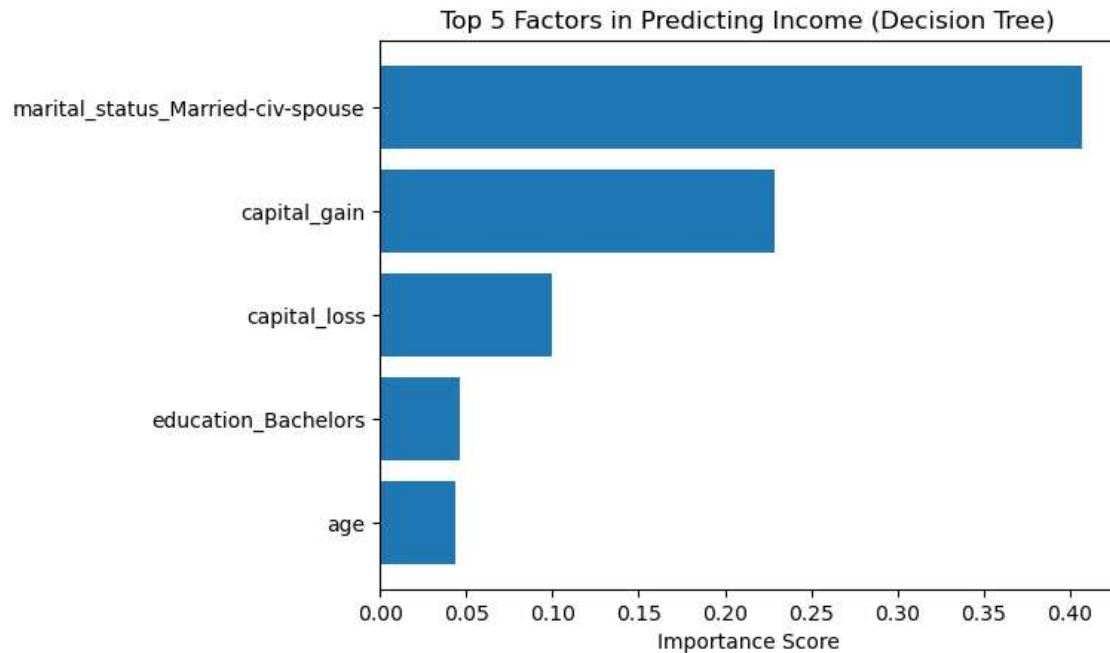


Figure 3: Factors Influencing Income (Decision Tree)

In contrast, Logistic Regression assigned capital gains as an important feature. Although the models assigned different weights to each feature, both identified the same two variables as the most influential predictors.

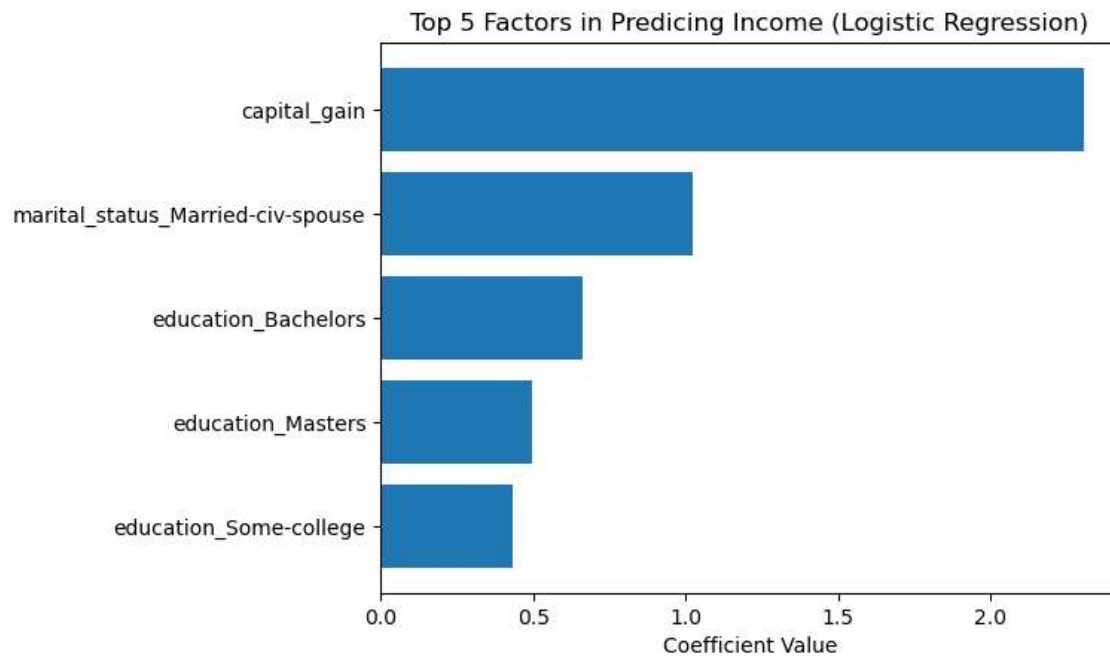


Figure 4: Factors Influencing Income (Decision Tree)

Interestingly, the Random Forest ensemble model did not demonstrate a notable improvement in predictive performance given its similar ROC-AUC performance of 0.9 compared to the. Thus, ensemble methods may be considered unnecessary in this context. Overall, all models achieved strong AUC scores, as shown in Figure 5, indicating good predictive ability despite the class imbalance.

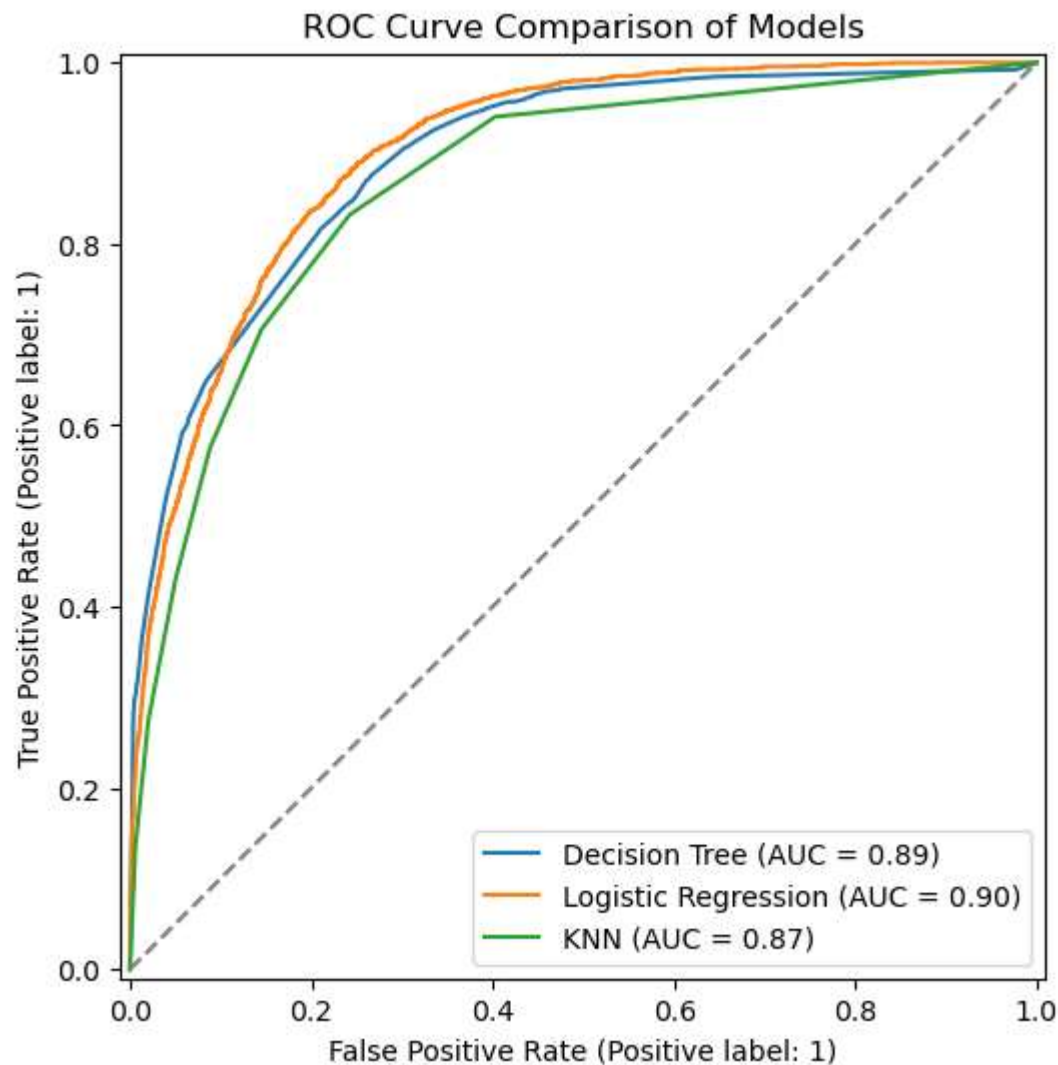


Figure 5: ROC Curve Comparison of Models

Discussion

The dataset performs well for classification tasks. However, feature selection and dimensionality reduction techniques may further improve model performance. Principal Component Analysis (PCA) is recommended, as one-hot encoding significantly increases the dimensionality of the

feature space. Reducing this expansion could enhance model performance. An alternative approach would be to collect numerical representations of features from the outset rather than relying heavily on categories. This may reduce sparsity and improve model performance. However, the current feature set already yields strong predictive results even with one-hot encoding.

The binary classification reduction of income may be limiting in terms of real-world interpretability. Income is inherently continuous, and multiclass labels (e.g., low, middle, and high income) may provide a more realistic and informative representation of socioeconomic predictors. Additionally, certain variables, such as race, raise ethical considerations. Including race in predictive modeling can introduce bias and lead to problematic interpretations. For this reason, such features should be excluded or carefully justified when used.

Conclusion

In conclusion, the classification models demonstrated strong and comparable performance in predicting income level, despite class imbalance. The results reinforce existing socioeconomic findings, particularly the importance of education and marital status as key predictors of income. Future research should use more recent census data to reflect current economic conditions. Lastly, exploring different machine learning models and multiclass income labels may provide more interpretable insights into income prediction.

References

Becker, B. & Kohavi, R. (1996). Adult [Dataset]. UCI Machine Learning Repository.
<https://doi.org/10.24432/C5XW20>.